

# **How to Solve Problems**

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# Acknowledgment

This is Evan Style LaTeX template. Thanks to Evan Chen for creating such a beautiful style file.

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1. Look at a certain statement or problem from different perspectives. For example: the following statements are equivalent:

$$p \mid t^n - 1$$

$t$  is a root of  $x^n - 1 \pmod{p}$

the order of  $t$  modulo  $p$  divides  $n$

2. Seek the relationship between objects and morphism. How to turn an object into a morphism and vice versa.

Given a group of automorphisms, we can look at the set of objects fixed by the m

Given a set of objects, we can look at the group of morphisms that fix them

An element in a field extension  $k$  to  $F$  can be seen as a linear transformation of  $F$

3. Look for the generalized version to dismiss the redundant information.
4. Create new definition and play with it to see how it fits into the problems.
5. Create extra objects and see how they interact with the existing objects.